THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Sound is produced by: 1

(a) stationary bodies **(b)** heat only
(c) light **(d)** vibrating bodies

2. The loudness of a sound depends mainly on its: 1

(a) speed **(b)** wavelength **(c)** amplitude **(d)** frequency

3. The minimum distance of the reflecting surface to hear an echo in air (v = 344 m/s) is about: 1

(a) 34.4 m **(b)** 17.2 m **(c)** 344 m **(d)** 1.72 m

4. Sound cannot travel through: 1

(a) steel **(b)** air **(c)** a vacuum **(d)** water

5. The pitch of a sound depends on its: 1

(a) medium **(b)** frequency **(c)** amplitude **(d)** speed

6. Bats find their prey in the dark by: 1

(a) light **(b)** infrasonic sounds
(c) echolocation using ultrasonic waves **(d)** smell

7. Sound waves in air are: 1

(a) light waves **(b)** transverse waves
(c) electromagnetic waves **(d)** longitudinal waves

8. The quality or timbre of a sound enables us to: 1

(a) find its loudness
(b) distinguish two sounds of the same pitch and loudness
(c) find its wavelength
(d) measure its speed

9. SONAR stands for: 1

(a) sound night and ranging **(b)** ship navigation and radar
(c) sound navigation and ranging **(d)** sound network and range

10. A region in a sound wave where the particles are crowded together is a: 1

(a) trough **(b)** compression **(c)** rarefaction **(d)** crest

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. How is sound produced? Name the medium needed for its propagation. 2

12. Distinguish between pitch and loudness. 2

13. State the audible range of frequencies of humans. What are infrasonic and ultrasonic sounds? 2

14. Why is sound called a longitudinal wave? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A wave has a frequency of 200 Hz and a wavelength of 1.7 m. Find its speed and time period. 3

16. Explain the working of SONAR and calculate the depth of the sea if the echo of an ultrasound pulse is received after 4 s (speed in water = 1500 m/s). 3

17. What is the effect of temperature on the speed of sound in air? Give a reason. 3

OR

Find the wavelengths in air (v = 344 m/s) for sound of frequencies 20 Hz and 20 kHz. 3

18. State three characteristics of a musical sound and the factor on which each depends. 3

OR

A sound of frequency 500 Hz travels in air at 340 m/s. Find the wavelength and the number of waves in 1 s and in 10 m. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the production and propagation of sound waves, with the definitions of compression, rarefaction, amplitude, wavelength, frequency and time period. 5

OR

Explain reverberation, its causes and effects in an auditorium, and the methods of controlling it, with the principle of curved sound boards. 5

20. Derive v = f lambda and use it to find the speed of sound in air if a wave has f = 256 Hz and lambda = 1.34 m. 5

OR

A ship sends an ultrasonic pulse, receives the echo after 2.4 s from the sea bed and the speed in sea water is 1500 m/s. Find the depth. If the bed is at 1650 m, find the echo time. 5

END OF QUESTION PAPER — Total: 40 Marks